ಬಿ.ಎಂ.ಎಸ್. ತಾಂತ್ರಿಕ ಮತ್ತು ವ್ಯವಸ್ಥಾಪನಾ ಮಹಾವಿದ್ಯಾಲಯ

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STAR FORGE

KAKSHA · satellite collision risk predictor: manoeuvre or monitor, two days out

TEAM NAME : **The Good Engineers**

TEAM LEADER : **Chirag**

TRACK : **Software**

PROBLEM STATEMENT CODE : **SW-5**





Problem and opportunity

The problem

• Satellites and debris pass at kilometres per second. Operators get conjunction data messages (CDMs) with a collision probability (Pc) that changes for days, and must decide early whether to manoeuvre.

• A manoeuvre costs fuel and mission time; a missed one can end a satellite (Iridium 33 and Cosmos 2251 collided in 2009).

• Dangerous events are rare: in ESA's held-out set, 150 of 2,167 events end high risk ($Pc \geq 10^{-6}$).

• The obvious rule, "trust the latest message", misses 35 of those 150 at two days out (our measurement on ESA data).

How KAKSHA is different

• A decision, not just a number: manoeuvre or monitor, with a calibrated probability, a range for the final risk and the three reasons behind it.

• Honest evaluation: ESA's real data and own score, split by event, nothing closer than 2 days; we reproduce ESA's published baseline score exactly.

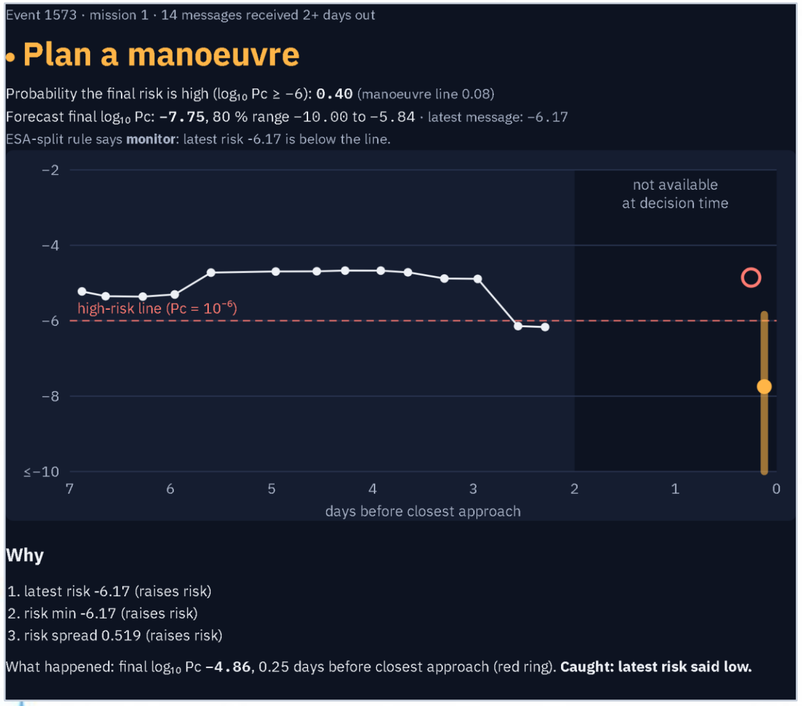
• The operator sets the fuel-against-risk line and sees what it costs.

USP: decisions you can check
ESA official split: L 0.676 vs 0.694 for the rule (small margin)
Retrained on recent events: 23 missed instead of 35,
for 37 more manoeuvres



Features

Real screenshots. Event 1573: latest message just under the line; KAKSHA said manoeuvre; final risk over 10x above it



1. Recommendation with confidence

Plan a manoeuvre or monitor, P(final risk high), and an 80 % range for the final risk.

2. Risk history with the 2-day cutoff

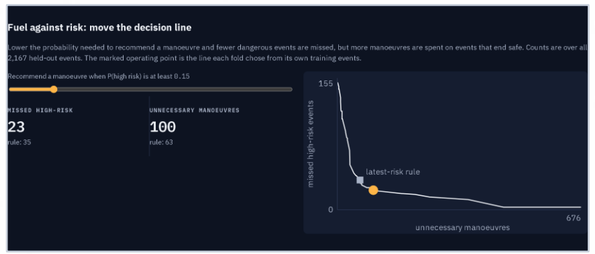
Every message for the event; the last two days are greyed out because they arrive too late.

3. Reasons

The three features that pushed this event's probability up or down.

4. Reveal and honest filters

Show what really happened; list our catches, our misses and our unnecessary manoeuvres.

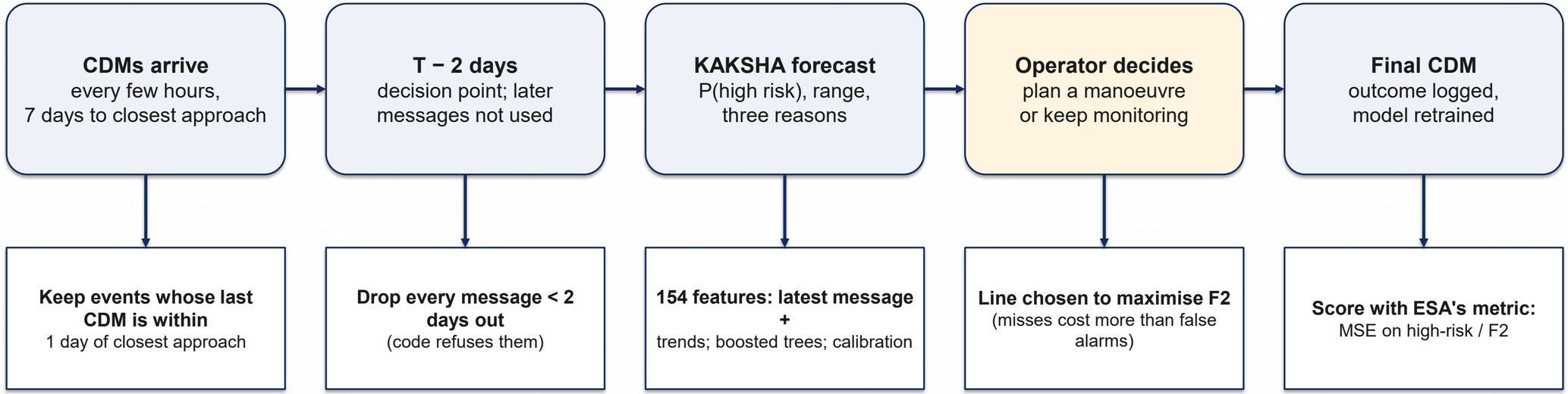


5. Fuel-against-risk slider

Move the line, see missed events against unnecessary manoeuvres.

Process flow: one close approach, from first warning to decision

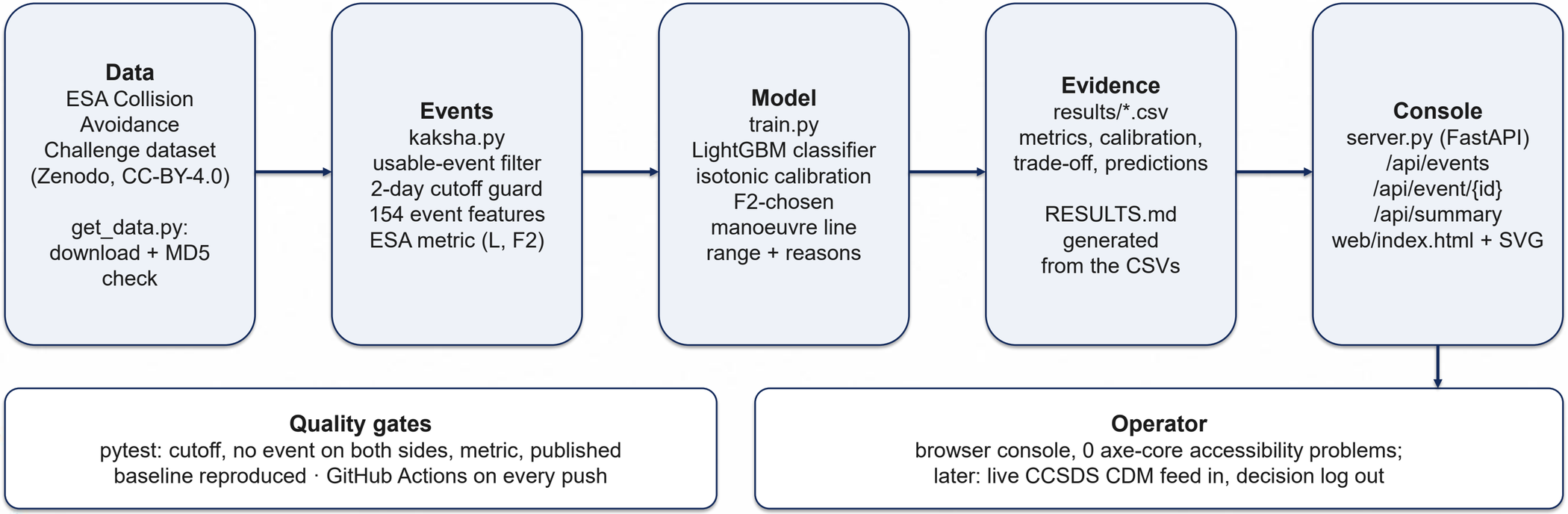
What the operator does with KAKSHA (top) and what KAKSHA does underneath (bottom)



Measured on 2,167 ESA held-out events: official split L 0.676 vs 0.694; retrained model misses 23 high-risk events where the latest-message rule misses 35. Use case: a flight-dynamics operator triaging dozens of close approaches a week, manoeuvring only where it matters.

Architecture

Every box exists and runs today (github.com/thegoodengineers/kaksha)



Modular on purpose: the model is one file behind a fixed feature table, so a sequence model can replace it without touching the console.

Technologies

Layer	What we use	Why
Language	Python 3.11	One language from data to API; fits a 24-hour build
Data	pandas, NumPy	103-column CDM tables, per-event grouping
Model	LightGBM gradient-boosted trees	Strong on tabular data, fast, gives per-event reasons
Calibration	scikit-learn isotonic regression	Probabilities that mean what they say
Uncertainty	Empirical (split-conformal style) range	No distribution assumed
API	FastAPI + Uvicorn	Three JSON endpoints, typed, quick
Console	Plain HTML, CSS, SVG charts	No framework; loads instantly; accessible
Quality	pytest, GitHub Actions, axe-core	Leakage, metric and accessibility checks on every push
Data source	ESA Kelvins dataset on Zenodo	Real CDMs, real labels, published baseline to beat



Extra 1: what we measured (ESA held-out events, two days out)

2,167 events, 150 high risk. ESA's score L = MSE on high-risk events / F2, lower is better.

1. ESA's official split

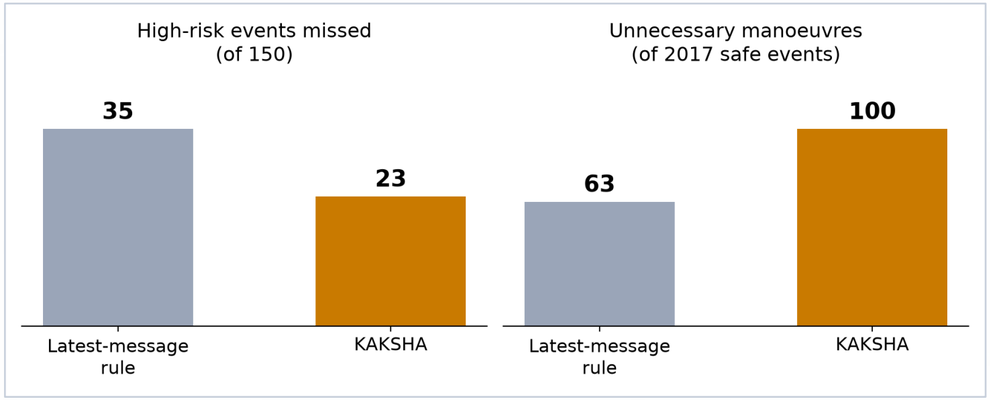
Rules tuned on ESA's train events only, scored on the test set once:

L 0.676 vs 0.694 for the latest-message rule

MSE 0.497 vs 0.513 · F2 0.736 vs 0.739 · missed 35 vs 35

Better in 75% of 2,000 bootstrap resamples: a small win, not yet certain.
Every test evaluation is logged (results/test_log.csv).

2. Retrained on recent events (LightGBM)



Said plainly

Our first model (LightGBM trained on ESA's train events) lost on the official split: L 0.849. ESA's train and test events differ: when the latest message said high risk, 261 of 308 train events ended low but only 63 of 178 test events, hence the retrained view (5 folds by event, no event scored by a model that saw it). The rules follow the published winner's approach; ESA's winner scored 0.556 and we have not matched it. Calibrated: Brier 0.032 vs 0.045. Ranges hold 70% of high-risk finals for an 80 % range: next to fix.

Extra 2: our 24-hour plan for Round 2

We start from a working, tested baseline, so the 24 hours go into making the model and the console better

Hours	What	Owner
H0–3	Data pipeline, re-run baseline and tests	Abhijeet
H3–10	Sequence features over the CDM history; per-mission calibration	Chirag
H10–16	Console: fleet view of all open events, decision log	Bhumika
H16–20	Evaluation: re-score, wider ranges where high-risk; cost slider in fuel terms	Abhijeet, Chirag
H20–23	Demo story (event 1573), pitch rehearsed twice with a timer	Ayush
H23–24	Buffer; recorded backup of the demo	Everyone

Rule for the 24 hours: every number we show comes from a script run that day.



References

- ESA, Collision Avoidance Challenge dataset, Zenodo, doi:10.5281/zenodo.4463683 (CC-BY-4.0).
- Uriot T., Izzo D., Simões L. F. et al., "Spacecraft collision avoidance challenge: design and results of a machine learning competition", Astrodynamics 6, 2022 (arXiv:2008.03069): rules, metric, published scores.
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- Zadrozny B., Elkan C., "Transforming classifier scores into accurate multiclass probability estimates", KDD 2002 (isotonic calibration).
- Lei J. et al., "Distribution-free predictive inference for regression", JASA 2018 (split-conformal ranges).
- CCSDS 508.0-B-1, Conjunction Data Message, Blue Book, 2013.
- NASA Orbital Debris Program Office, Orbital Debris Quarterly News 13(2), April 2009 (Iridium 33 / Cosmos 2251 collision).

Code, results and tests: github.com/thegoodengineers/kaksha · Team: The Good Engineers (Chirag, Bhumika, Ayush, Abhijeet)

